

Whole Blood Smear

Advanced AI + Explainable AI Analysis

Cellpose • ResNet50 • Grad-CAM • SHAP • Vision-Language Analysis

Purpose

This report documents a research-oriented computer-vision analysis of a whole blood-smear image. The system segments candidate objects, classifies individual cells using ResNet50, and analyzes selected predictions using Grad-CAM and SHAP.

What makes this report different

Instead of presenting only classification scores and heatmaps, this report analyzes the relationship between the model prediction, the second-best class, prediction margin, segmentation measurements, Grad-CAM, SHAP, and potential model limitations.

Important research statement

The output is a computational screening and model-evaluation result. It is not a diagnosis of cancer, leukemia, AML, or any other disease.

Analysis date: 07 September 2026

1. Executive Summary

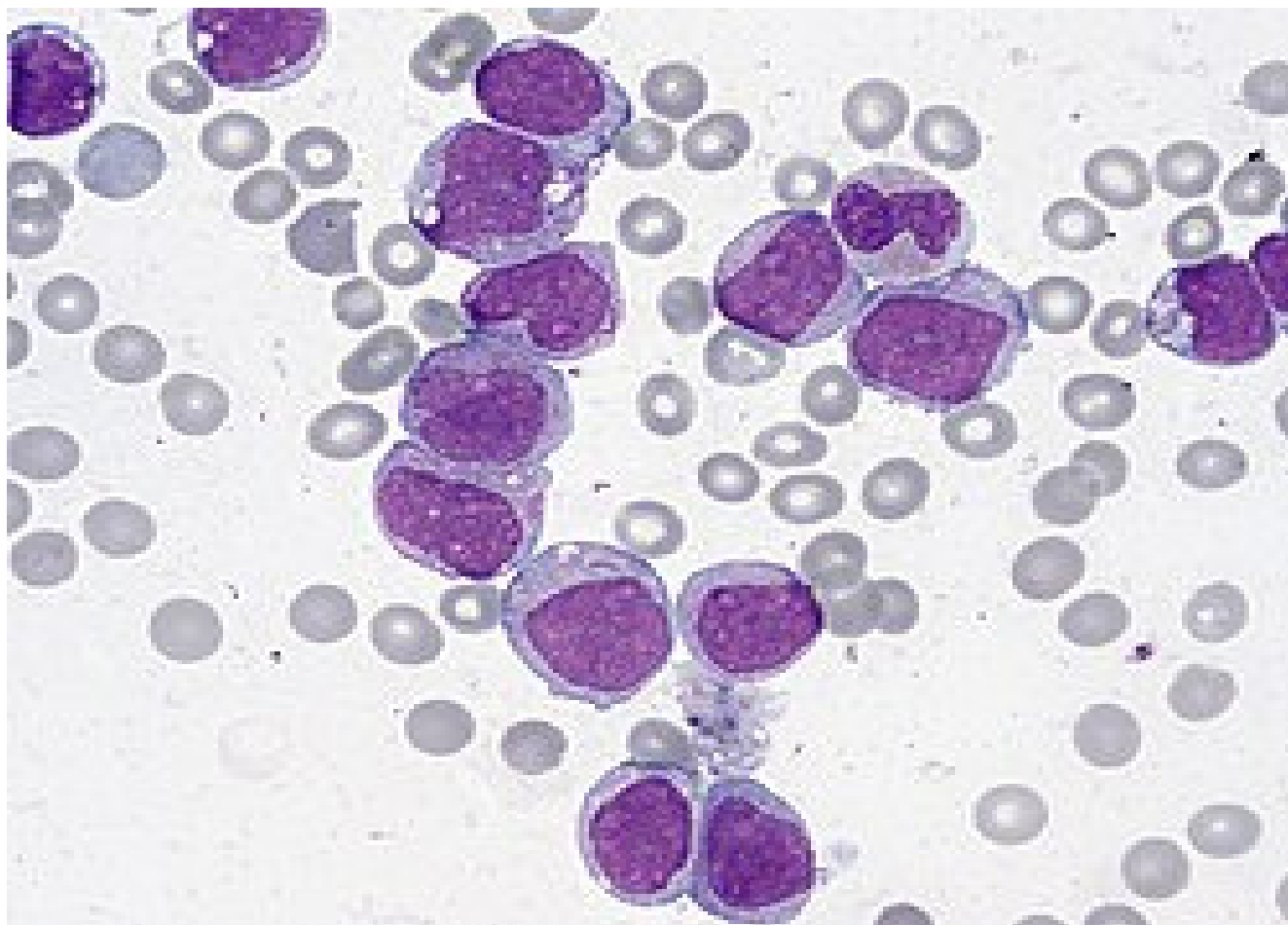
The segmentation stage identified **83** Cellpose objects. After the existing purple-score candidate filter, **16** objects were retained for cell-level classification. Among these candidates, **9** received a myeloblast prediction. This corresponds to **56.25%** of the candidate objects analyzed by this particular computational filter. The myeloblast predictions had model confidence values ranging from **65.8%** to **93.3%**, with an average of **81.6%**. These proportions describe the output of the current segmentation and classification pipeline. They should not be interpreted as the percentage of leukemic cells in a patient or as a cancer probability.

Pipeline Stage	Result
Cellpose segmentation	83 objects
Candidate filtering	16 candidates
Myeloblast-like predictions	9
Candidate myeloblast proportion	56.25%
Average candidate confidence	84.10%

Interpretation: The reported myeloblast-like proportion is a proportion of objects selected by this computational candidate-selection procedure. It is not a clinical blast percentage and should not be used as a standalone diagnostic threshold.

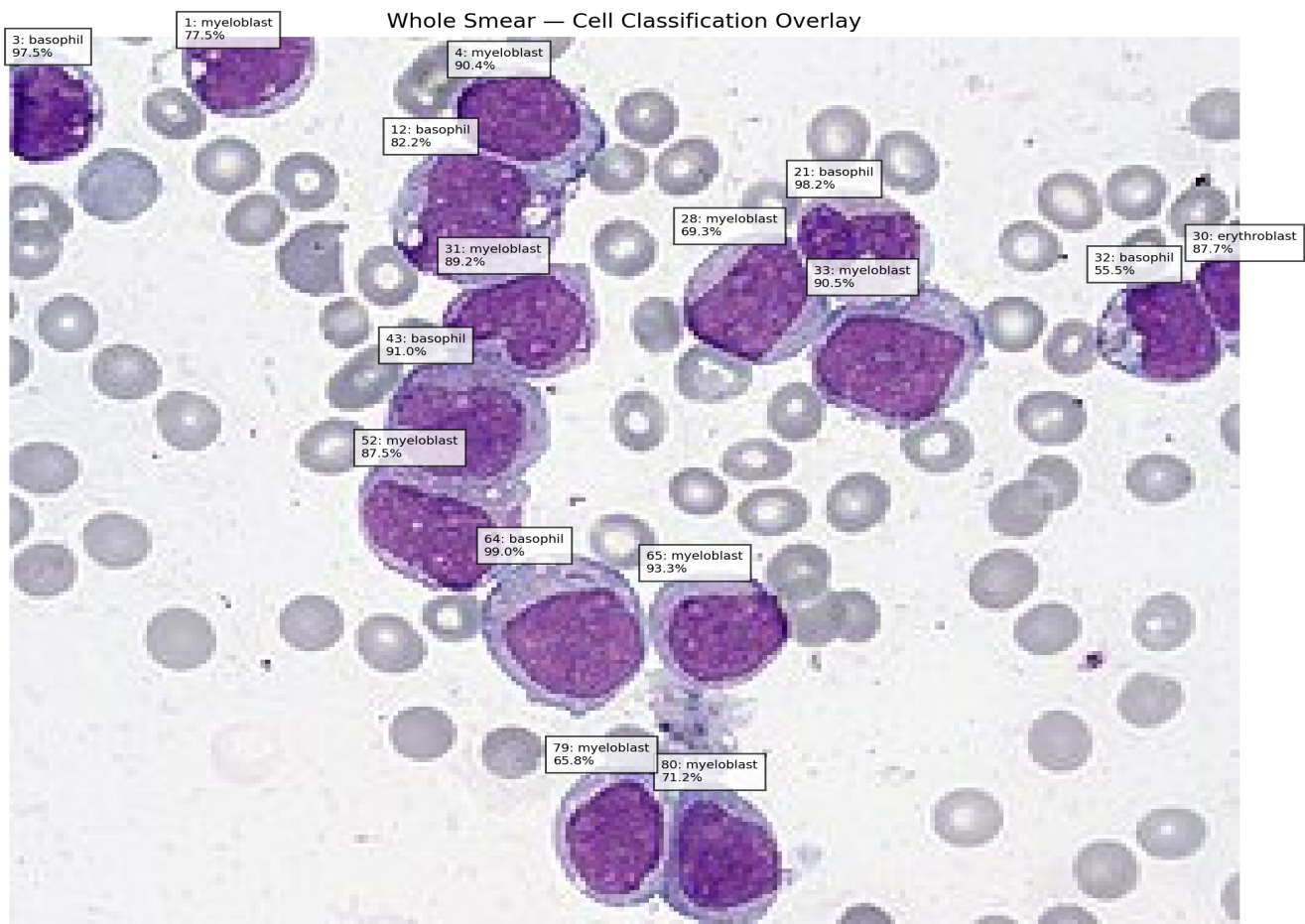
2. Whole-Smear Context

The whole-smear image provides the original microscopic context before individual objects are evaluated by the classifier.



Interpretation: At this stage the system has not yet established biological identity. The image is the source from which Cellpose creates candidate object masks.

3. Segmentation and Classification Map



The overlay associates candidate objects with the five available ResNet50 classes. The labels represent the classifier's predictions, not independent expert annotations.

Important model limitation: The classifier has five available outputs and does not contain an explicit mature-RBC or unknown-cell category. An object outside the training distribution can therefore still receive a high-confidence prediction.

4. Candidate Population Analysis

The following table describes the distribution of ResNet50 predictions among the objects that passed the existing candidate-selection rule.

Predicted Class	Candidates	Percentage	Mean Confidence
basophil	6	37.50%	87.21%
erythroblast	1	6.25%	87.73%
monocyte	0	0.00%	0.00%
myeloblast	9	56.25%	81.63%
seg_neutrophil	0	0.00%	0.00%

Why this matters: Looking at the distribution rather than a single prediction provides a more useful description of the model's behavior across the smear.

5. Detailed Cell-Level Results

Cell	Prediction	Confidence	2nd Class	2nd Score	Margin
65	myeloblast	93.3%	erythroblast	3.7%	89.6%
33	myeloblast	90.5%	basophil	5.7%	84.8%
4	myeloblast	90.4%	basophil	9.3%	81.0%
31	myeloblast	89.2%	erythroblast	5.6%	83.6%
52	myeloblast	87.5%	monocyte	6.8%	80.7%
1	myeloblast	77.5%	basophil	20.2%	57.3%

New metric — prediction margin: The margin is the difference between the highest and second-highest class probability. It is useful because two cells can both have high confidence while having very different separation from the competing class.

6. Deep Explainable-AI Analysis

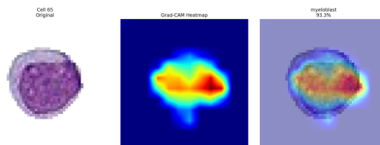
The following sections combine the original cell image, Grad-CAM, SHAP, classification probabilities and a detailed interpretation of the model evidence.

6.1 Cell 65 — myeloblast (93.3%)

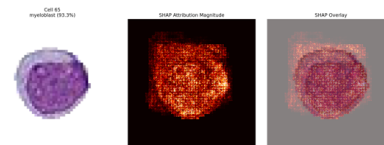
ORIGINAL CELL



GRAD-CAM



SHAP



Metric	Value
Prediction	myeloblast
Model confidence	93.28%
Second-best class	erythroblast
Second-best score	3.65%
Prediction margin	89.62%
Segmented area	631 px
Purple score	50.25
Grad-CAM	Available
SHAP	Available

MODEL DECISION

For Cell 65, the ResNet50 classifier selected **myeloblast** with a model confidence of **93.28%**. The second-highest class was **erythroblast** at **3.65%**.

The model selected this class with a strong softmax preference.

The predicted class exceeds the second-best class by approximately 89.6 percentage points.

This describes the model's internal probability distribution.
It should not be interpreted as a clinical probability.

XAI EVIDENCE

Grad-CAM and SHAP were generated specifically for the selected **myeloblast** output. Grad-CAM provides a spatial

indication of regions contributing to the network's prediction, while SHAP provides pixel-level attribution magnitude.

The available XAI visualizations should therefore be interpreted as evidence about which image regions influenced the classifier, rather than as direct measurements of cell morphology.

The segmentation area for this cell is approximately **631 pixels**, with a purple-score of approximately **50.2**.

CROSS-METHOD CONSISTENCY

Both Grad-CAM and SHAP are available for this cell.

When the two attribution methods emphasize similar parts of the segmented cell, this provides stronger evidence that the model is using internally consistent image regions. When their patterns differ substantially, the prediction should be interpreted more cautiously because different explanatory methods may reveal different aspects of the network's decision.

RELIABILITY INTERPRETATION

The confidence score indicates how strongly the five-class model preferred the selected class relative to its other available outputs. It does not measure diagnostic certainty.

This distinction is particularly important when analyzing images that differ from the training dataset in staining, illumination, microscope characteristics, cell preparation, or biological appearance.

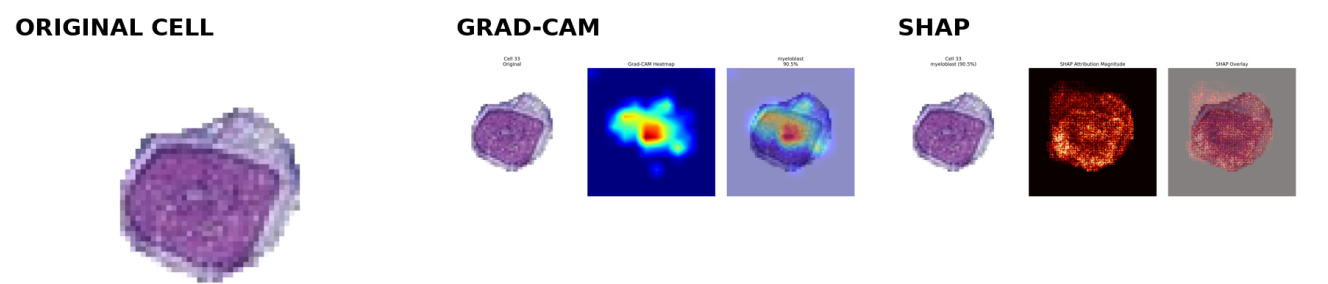
SCIENTIFIC LIMITATION

This explanation describes computational model behavior only. Grad-CAM and SHAP do not establish that the predicted class is biologically correct, and they do not independently establish the presence or absence of leukemia or another disease.

The current classifier contains five classes and does not contain an explicit mature-RBC or unknown-cell category. Consequently, objects outside the training distribution can still receive a high-confidence prediction among the available classes.

Evidence interpretation note: Grad-CAM and SHAP explain the behavior of the trained model. They do not provide independent biological confirmation of the predicted class.

6.2 Cell 33 — myeloblast (90.5%)



Metric	Value
Prediction	myeloblast
Model confidence	90.47%
Second-best class	basophil
Second-best score	5.66%
Prediction margin	84.80%
Segmented area	895 px
Purple score	47.40
Grad-CAM	Available
SHAP	Available

MODEL DECISION

For Cell 33, the ResNet50 classifier selected **myeloblast** with a model confidence of **90.47%**. The second-highest class was **basophil** at **5.66%**.

The model selected this class with a strong softmax preference.

The predicted class exceeds the second-best class by approximately 84.8 percentage points.

This describes the model's internal probability distribution.
It should not be interpreted as a clinical probability.

XAI EVIDENCE

Grad-CAM and SHAP were generated specifically for the selected **myeloblast** output. Grad-CAM provides a spatial indication of regions contributing to the network's prediction, while SHAP provides pixel-level attribution magnitude.

The available XAI visualizations should therefore be interpreted

as evidence about which image regions influenced the classifier, rather than as direct measurements of cell morphology.

The segmentation area for this cell is approximately **895 pixels**, with a purple-score of approximately **47.4**.

CROSS-METHOD CONSISTENCY

Both Grad-CAM and SHAP are available for this cell.

When the two attribution methods emphasize similar parts of the segmented cell, this provides stronger evidence that the model is using internally consistent image regions. When their patterns differ substantially, the prediction should be interpreted more cautiously because different explanatory methods may reveal different aspects of the network's decision.

RELIABILITY INTERPRETATION

The confidence score indicates how strongly the five-class model preferred the selected class relative to its other available outputs. It does not measure diagnostic certainty.

This distinction is particularly important when analyzing images that differ from the training dataset in staining, illumination, microscope characteristics, cell preparation, or biological appearance.

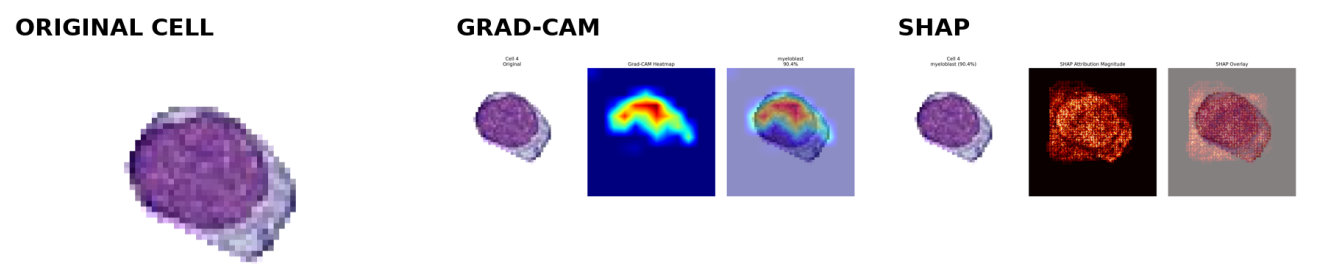
SCIENTIFIC LIMITATION

This explanation describes computational model behavior only. Grad-CAM and SHAP do not establish that the predicted class is biologically correct, and they do not independently establish the presence or absence of leukemia or another disease.

The current classifier contains five classes and does not contain an explicit mature-RBC or unknown-cell category. Consequently, objects outside the training distribution can still receive a high-confidence prediction among the available classes.

Evidence interpretation note: Grad-CAM and SHAP explain the behavior of the trained model. They do not provide independent biological confirmation of the predicted class.

6.3 Cell 4 — myeloblast (90.4%)



Metric	Value
Prediction	myeloblast
Model confidence	90.38%
Second-best class	basophil
Second-best score	9.33%
Prediction margin	81.05%
Segmented area	598 px
Purple score	52.03
Grad-CAM	Available
SHAP	Available

MODEL DECISION

For Cell 4, the ResNet50 classifier selected **myeloblast** with a model confidence of **90.38%**. The second-highest class was **basophil** at **9.33%**.

The model selected this class with a strong softmax preference.

The predicted class exceeds the second-best class by approximately 81.0 percentage points.

This describes the model's internal probability distribution.
It should not be interpreted as a clinical probability.

XAI EVIDENCE

Grad-CAM and SHAP were generated specifically for the selected **myeloblast** output. Grad-CAM provides a spatial indication of regions contributing to the network's prediction, while SHAP provides pixel-level attribution magnitude.

The available XAI visualizations should therefore be interpreted

as evidence about which image regions influenced the classifier, rather than as direct measurements of cell morphology.

The segmentation area for this cell is approximately **598 pixels**, with a purple-score of approximately **52.0**.

CROSS-METHOD CONSISTENCY

Both Grad-CAM and SHAP are available for this cell.

When the two attribution methods emphasize similar parts of the segmented cell, this provides stronger evidence that the model is using internally consistent image regions. When their patterns differ substantially, the prediction should be interpreted more cautiously because different explanatory methods may reveal different aspects of the network's decision.

RELIABILITY INTERPRETATION

The confidence score indicates how strongly the five-class model preferred the selected class relative to its other available outputs. It does not measure diagnostic certainty.

This distinction is particularly important when analyzing images that differ from the training dataset in staining, illumination, microscope characteristics, cell preparation, or biological appearance.

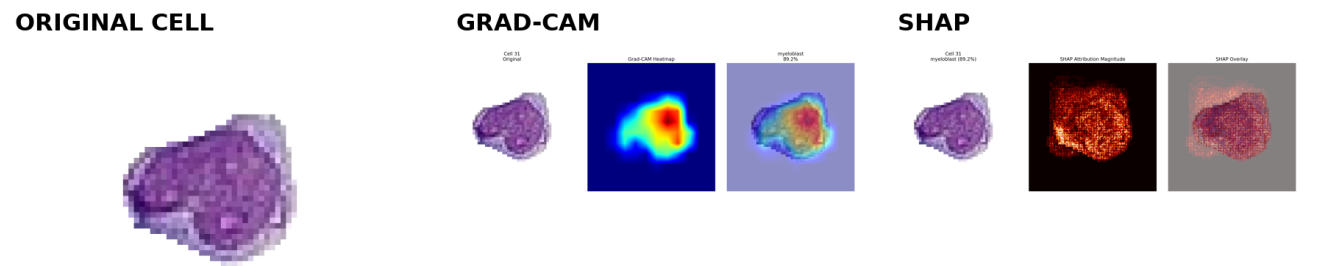
SCIENTIFIC LIMITATION

This explanation describes computational model behavior only. Grad-CAM and SHAP do not establish that the predicted class is biologically correct, and they do not independently establish the presence or absence of leukemia or another disease.

The current classifier contains five classes and does not contain an explicit mature-RBC or unknown-cell category. Consequently, objects outside the training distribution can still receive a high-confidence prediction among the available classes.

Evidence interpretation note: Grad-CAM and SHAP explain the behavior of the trained model. They do not provide independent biological confirmation of the predicted class.

6.4 Cell 31 — myeloblast (89.2%)



Metric	Value
Prediction	myeloblast
Model confidence	89.19%
Second-best class	erythroblast
Second-best score	5.64%
Prediction margin	83.55%
Segmented area	647 px
Purple score	52.83
Grad-CAM	Available
SHAP	Available

MODEL DECISION

For Cell 31, the ResNet50 classifier selected **myeloblast** with a model confidence of **89.19%**. The second-highest class was **erythroblast** at **5.64%**.

The model showed a substantial preference for this class, although the prediction is less separated from alternative classes than a very high-confidence prediction.

The predicted class exceeds the second-best class by approximately 83.6 percentage points.

This describes the model's internal probability distribution. It should not be interpreted as a clinical probability.

XAI EVIDENCE

Grad-CAM and SHAP were generated specifically for the selected **myeloblast** output. Grad-CAM provides a spatial indication of regions contributing to the network's prediction, while SHAP provides pixel-level attribution magnitude.

The available XAI visualizations should therefore be interpreted as evidence about which image regions influenced the classifier, rather than as direct measurements of cell morphology.

The segmentation area for this cell is approximately **647 pixels**, with a purple-score of approximately **52.8**.

CROSS-METHOD CONSISTENCY

Both Grad-CAM and SHAP are available for this cell.

When the two attribution methods emphasize similar parts of the segmented cell, this provides stronger evidence that the model is using internally consistent image regions. When their patterns differ substantially, the prediction should be interpreted more cautiously because different explanatory methods may reveal different aspects of the network's decision.

RELIABILITY INTERPRETATION

The confidence score indicates how strongly the five-class model preferred the selected class relative to its other available outputs. It does not measure diagnostic certainty.

This distinction is particularly important when analyzing images that differ from the training dataset in staining, illumination, microscope characteristics, cell preparation, or biological appearance.

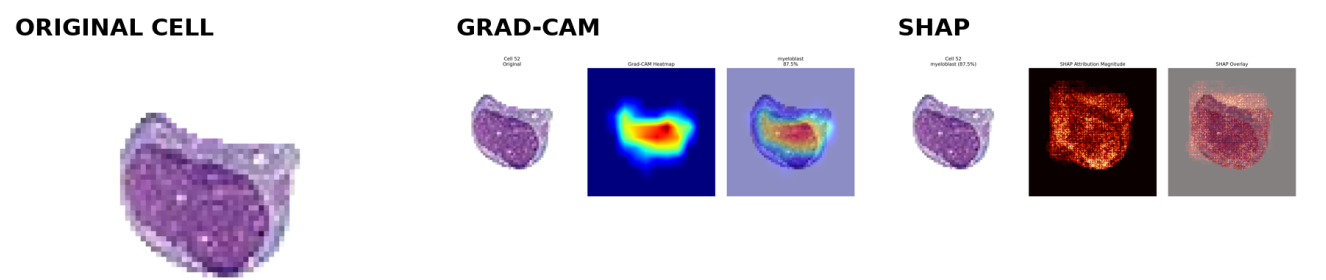
SCIENTIFIC LIMITATION

This explanation describes computational model behavior only. Grad-CAM and SHAP do not establish that the predicted class is biologically correct, and they do not independently establish the presence or absence of leukemia or another disease.

The current classifier contains five classes and does not contain an explicit mature-RBC or unknown-cell category. Consequently, objects outside the training distribution can still receive a high-confidence prediction among the available classes.

Evidence interpretation note: Grad-CAM and SHAP explain the behavior of the trained model. They do not provide independent biological confirmation of the predicted class.

6.5 Cell 52 — myeloblast (87.5%)



Metric	Value
Prediction	myeloblast
Model confidence	87.53%
Second-best class	monocyte
Second-best score	6.79%
Prediction margin	80.75%
Segmented area	822 px
Purple score	48.80
Grad-CAM	Available
SHAP	Available

MODEL DECISION

For Cell 52, the ResNet50 classifier selected **myeloblast** with a model confidence of **87.53%**. The second-highest class was **monocyte** at **6.79%**.

The model showed a substantial preference for this class, although the prediction is less separated from alternative classes than a very high-confidence prediction.

The predicted class exceeds the second-best class by approximately 80.7 percentage points.

This describes the model's internal probability distribution. It should not be interpreted as a clinical probability.

XAI EVIDENCE

Grad-CAM and SHAP were generated specifically for the selected **myeloblast** output. Grad-CAM provides a spatial indication of regions contributing to the network's prediction, while SHAP provides pixel-level attribution magnitude.

The available XAI visualizations should therefore be interpreted as evidence about which image regions influenced the classifier, rather than as direct measurements of cell morphology.

The segmentation area for this cell is approximately **822 pixels**, with a purple-score of approximately **48.8**.

CROSS-METHOD CONSISTENCY

Both Grad-CAM and SHAP are available for this cell.

When the two attribution methods emphasize similar parts of the segmented cell, this provides stronger evidence that the model is using internally consistent image regions. When their patterns differ substantially, the prediction should be interpreted more cautiously because different explanatory methods may reveal different aspects of the network's decision.

RELIABILITY INTERPRETATION

The confidence score indicates how strongly the five-class model preferred the selected class relative to its other available outputs. It does not measure diagnostic certainty.

This distinction is particularly important when analyzing images that differ from the training dataset in staining, illumination, microscope characteristics, cell preparation, or biological appearance.

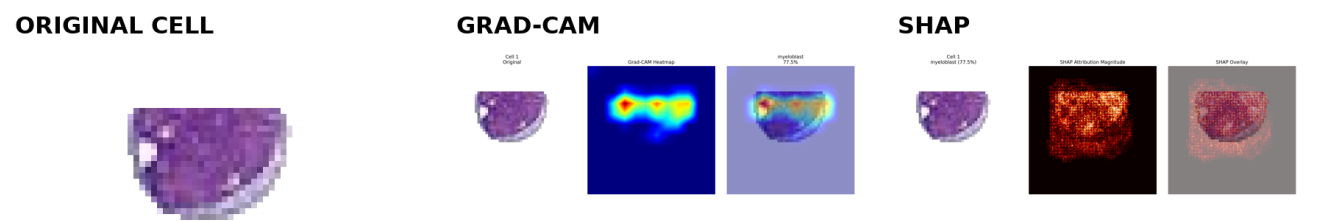
SCIENTIFIC LIMITATION

This explanation describes computational model behavior only. Grad-CAM and SHAP do not establish that the predicted class is biologically correct, and they do not independently establish the presence or absence of leukemia or another disease.

The current classifier contains five classes and does not contain an explicit mature-RBC or unknown-cell category. Consequently, objects outside the training distribution can still receive a high-confidence prediction among the available classes.

Evidence interpretation note: Grad-CAM and SHAP explain the behavior of the trained model. They do not provide independent biological confirmation of the predicted class.

6.6 Cell 1 — myeloblast (77.5%)



Metric	Value
Prediction	myeloblast
Model confidence	77.50%
Second-best class	basophil
Second-best score	20.21%
Prediction margin	57.29%
Segmented area	446 px
Purple score	53.46
Grad-CAM	Available
SHAP	Available

MODEL DECISION

For Cell 1, the ResNet50 classifier selected **myeloblast** with a model confidence of **77.50%**. The second-highest class was **basophil** at **20.21%**.

The model showed a substantial preference for this class, although the prediction is less separated from alternative classes than a very high-confidence prediction.

The predicted class exceeds the second-best class by approximately 57.3 percentage points.

This describes the model's internal probability distribution.
It should not be interpreted as a clinical probability.

XAI EVIDENCE

Grad-CAM and SHAP were generated specifically for the selected **myeloblast** output. Grad-CAM provides a spatial indication of regions contributing to the network's prediction, while SHAP provides pixel-level attribution magnitude.

The available XAI visualizations should therefore be interpreted as evidence about which image regions influenced the classifier, rather than as direct measurements of cell morphology.

The segmentation area for this cell is approximately **446 pixels**, with a purple-score of approximately **53.5**.

CROSS-METHOD CONSISTENCY

Both Grad-CAM and SHAP are available for this cell.

When the two attribution methods emphasize similar parts of the segmented cell, this provides stronger evidence that the model is using internally consistent image regions. When their patterns differ substantially, the prediction should be interpreted more cautiously because different explanatory methods may reveal different aspects of the network's decision.

RELIABILITY INTERPRETATION

The confidence score indicates how strongly the five-class model preferred the selected class relative to its other available outputs. It does not measure diagnostic certainty.

This distinction is particularly important when analyzing images that differ from the training dataset in staining, illumination, microscope characteristics, cell preparation, or biological appearance.

SCIENTIFIC LIMITATION

This explanation describes computational model behavior only. Grad-CAM and SHAP do not establish that the predicted class is biologically correct, and they do not independently establish the presence or absence of leukemia or another disease.

The current classifier contains five classes and does not contain an explicit mature-RBC or unknown-cell category. Consequently, objects outside the training distribution can still receive a high-confidence prediction among the available classes.

Evidence interpretation note: Grad-CAM and SHAP explain the behavior of the trained model. They do not provide independent biological confirmation of the predicted class.

7. Cross-Cell Comparison

Examining several predictions together is important because an individual high-confidence prediction does not reveal whether the model behaves consistently across different cells.

The highest-confidence selected prediction was **Cell 65**, classified as **myeloblast** at **93.28%**.

The lowest-confidence selected prediction was **Cell 1**, classified as **myeloblast** at **77.50%**.

The widest prediction margin occurred for Cell **65** (89.62 percentage points).

The narrowest prediction margin occurred for Cell **1** (57.29 percentage points), making it the most ambiguous of the selected predictions according to this metric.

Why this is useful: A model can have high confidence while still being wrong when the input is outside its training distribution. Prediction margins, XAI maps and cross-cell consistency therefore provide additional information for evaluating model behavior.

8. Model Reliability and Domain-Shift Analysis

A major consideration for this experiment is the difference between the data used to train the ResNet50 model and the appearance of cells in an external whole-smear image.

Training-domain behavior: The model learned a five-class classification problem. It therefore expects every input to resemble one of those five categories.

Closed-set behavior: If an object is actually an RBC, lymphocyte, artifact, segmentation fragment, or another cell type outside the five training classes, the network still has to choose one of its five outputs.

Why confidence is not enough: Softmax confidence measures preference among the available classes. It does not tell us whether the input truly belongs to the training distribution.

Why XAI matters: If Grad-CAM and SHAP frequently emphasize boundaries, background regions, staining artifacts, or other non-cellular features, this may indicate that the classifier is using a shortcut rather than a robust biological feature.

Recommended research interpretation: The system should be evaluated as a cell-level screening and classification pipeline rather than as a standalone cancer detector.

9. What These Results Do — and Do Not — Mean

What the system can report:

- Which of the five trained classes the model selected for a segmented object.
- How strongly the model preferred that class relative to its other outputs.
- Which image regions are emphasized by Grad-CAM.
- Which pixels contribute strongly to the SHAP attribution.
- How predictions are distributed across the selected candidate objects.
- Where model behavior appears consistent or potentially unreliable.

What the system cannot establish by itself:

- A diagnosis of cancer.
- A diagnosis of AML.
- That every predicted myeloblast is biologically a myeloblast.
- That the myeloblast-like proportion is a clinical blast percentage.
- That model confidence represents clinical probability.
- That Grad-CAM or SHAP proves a specific biological structure.

10. Recommended Next Development

Add an explicit RBC / background class

This would reduce the closed-set problem in whole-smear images.

Add an unknown / reject option

Objects that do not resemble the training distribution should be allowed to remain unclassified.

Validate on external cell-level datasets

Evaluation should include images from different staining protocols, microscopes and acquisition conditions.

Compare Grad-CAM and SHAP systematically

Rather than inspecting only individual examples, quantify whether attribution remains inside segmented cell regions.

Evaluate cell-level sensitivity and specificity

Medical screening performance should not be summarized using accuracy alone.

Use expert annotations for external validation

Whole-image captions should not be treated as labels for every individual cell.

Keep the VLM as an explanation layer

The language model should organize and explain evidence generated by the validated vision pipeline rather than independently diagnosing the smear.

11. Conclusion

The resulting pipeline combines whole-smear segmentation, five-class ResNet50 classification and two complementary explainability methods. The additional evidence analysis provides a more complete picture of how the model behaves on individual cells.

In this analysis, 83 objects were segmented and 16 passed the existing candidate-selection procedure. 9 candidate objects received a myeloblast prediction, corresponding to 56.25% of the selected candidate population.

These numbers describe the computational pipeline and should not be interpreted as a patient-level cancer probability or as a clinical blast count.

The most valuable contribution of the XAI stage is not simply producing colorful heatmaps. It is determining whether the model's decision is supported by consistent image evidence and whether the attribution exposes potential shortcuts, background dependence or domain-shift problems.

Final research statement: The system is best described as an AI-assisted blood-cell screening and explainability pipeline. Its predictions require independent hematopathological validation before any clinical interpretation.

12. Technical Appendix

Component	Configuration
Classifier	ResNet50
Input resolution	384 × 384
Number of classes	5
Classes	basophil, erythroblast, monocyte, myeloblast, seg_neutrophil
Segmentation	Cellpose
Grad-CAM target layer	conv5_block3_out
Explainability methods	Grad-CAM and SHAP
Language explanation layer	Qwen2.5-VL-3B-Instruct when available
Report generator	ReportLab

Report generation status: Qwen2.5-VL explanations were generated.